KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
Nurturing Every Learner's Potential

SENIOR SCHOOL CURRICULUM DESIGN

GRADE 10

PHYSICS

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FOREWORD

The Government of Kenya is committed to ensuring that policy objectives for Education, Training and Research meet the provisions of the Kenya Constitution 2010 and Vision 2030, East African Community (EAC) Protocol, African Agenda 2063, United Nations Sustainable Development Goals (SDGs) and demands of the 21st Century.

Towards achieving the vision and mission of Basic Education, the Ministry of Education (MoE) has successfully and progressively rolled out the Competency Based Curriculum (CBC) at Pre-Primary (PP1& PP2), Primary (Grades 1-6) and Junior School (Grades 7-9) levels of education. The first cohort of CBC learners will transit to Senior School (Grades 10-12) in 2026.

The curriculum designs at this level facilitate learners to acquire relevant knowledge, skills, attitudes and values in preparation for Tertiary Education and the world of work.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

I urge all Government agencies and other stakeholders in Education to use the designs to plan for effective and efficient implementation of the Competency Based Curriculum.

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PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Curriculum (CBC) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017 during the national convention in January 2017. The CBC was piloted in 2017 and 2018 and rolled out at Pre-Primary, Primary and Grades 1 to 3 in 2019. The curriculum was then introduced in the subsequent grades through a phase in - phase out strategy, to ensure a smooth transition. Currently, the first Cohort of CBC are at Grade 9 – the last grade at Junior School. The implementation of CBC has also benefited immensely from the recommendations of the Presidential Working Party on Education Reforms (PWPER) that were released in

August 2023 and fully implemented, in terms of Curriculum design and curriculum support materials.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways and tracks of their choice, guided by their potentials and interests as demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and/or development of initial competencies for the Industry. This is an important step in the realisation of the Vision of the curriculum reforms; which is to produce *engaged, empowered and ethical citizens*. The vision and mission of CBC as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs indicate, among others, suggested learning experiences and resources. The Ministry of Education, through the State Department for Basic Education commits to provide these resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers and other officers involved in supporting and quality assuring curriculum implementation to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 mandates the Institute to develop curricula and curriculum support materials for Basic and Tertiary Education. The curriculum development process involves thorough research, and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD,2017).

KICD receives its funding from the Government of Kenya (GOK) to enable the successful achievement of its mandate. The Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank through the Kenya Primary Education Equity in Learning Programme (KPEELP); a project coordinated by MoE. Therefore, the Institute is very grateful to the GOK, through the MoE and the development partners for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also acknowledge the KICD staff, all teachers, educators who took part as panelists and various stakeholders. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC)and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide implementation of the CBC at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. Foster nationalism and patriotism and promote national unity.

Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. Promote the social, economic, technological and industrial needs for national development.

Education should prepare the youth of the country to play an effective and productive role in the life of the nation. **a) Social Needs**

Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution in the wake of rapid modernisation. Education should assist our youth to adapt to this change.

b) Economic Needs

Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building a modern and independent economy which is in need of an adequate and relevant domestic workforce.

c) Technological and Industrial Needs

Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognises the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

3. Promote individual development and self-fulfilment

Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

4. Promote sound moral and religious values.

Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

5. Promote social equity and responsibility.

Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. Promote respect for and development of Kenya's rich and varied cultures.

Education should instill in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.

7. Promote international consciousness and foster positive attitudes towards other nations.

Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

8. Promote positive attitudes towards good health and environmental protection.

Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that learners will transition to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre university/ pre-career experience in which the learners have an opportunity to choose pathways and tracks where they shall have demonstrated interest and potential at the earlier levels. Senior School comprises three years of education for learners in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be *engaged, empowered and ethical citizens* ready to participate in the socio-economic development of the nation.

At this level, learners shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise four core learning areas namely: English, Kiswahili, Community Service Learning (CSL) and Physical Education (PE). The learner will select three learning areas depending on career choice, aptitude, interest and personality and with guidance by the career teacher at Senior School.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:
1. Communicate effectively and utilise information and communication technology across varied contexts.
2. Apply mathematical, logical and critical thinking skills for problem solving.
3. Apply basic research and scientific skills to manipulate the environment and solve problems. 4.

Exploit individual talents for leisure, self-fulfilment, career growth, further education and training. 5. Uphold national, moral and religious values and apply them in day-to-day life.

6. Apply and promote health care strategies in day-to-day life.

7. Protect, preserve and improve the environment for sustainability.

8. Demonstrate active local and global citizenship for harmonious co-existence.

9. Demonstrate appreciation of diversity in people and cultures.

10. Manage pertinent and contemporary issues responsibly.

LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)

1. English 2. Kiswahili/KSL 3. Community Service Learning (CSL) 4. Physical Education (PE)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Mathematics 24. Biology 25. Chemistry 26. Physics 27. General Science 28. Agriculture 29. Computer Studies 30. Home Science 31. Aviation 32. Building Construction 33. Electricity 34. Metalwork 35. Power Mechanics 36. Woodwork 37. Media Technology 38. Marine and Fisheries Technology

LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	**No of Lessons Per Week (40 minutes per lesson)**
Core Subjects 4	

1. English	6
2. Kiswahili	6
3. Physical Education (PE)	3
4. Community Service Learning (CSL)	3
Elective Subjects	
5. Elective 1	6
6. Elective 2	6
7. Elective 3	6
** ICT Skills*	*2*
**PPI*	*1*
** Learner Personal /Group Study*	*1*
Total Number of Lessons	**40**

***Note:**

a) ICT skills will be offered to all learners to facilitate learning and enjoyment. These skills include accessing, creating, digital citizenship, and caring for ICT devices. The teacher should use ICT skills for delivery of lessons. b) Every school shall offer Pastoral Programme Instruction (PPI) for moral, spiritual and character development of the learners.

ESSENCE STATEMENT

Physics is a body of knowledge exploring the theories, principles and laws that govern natural phenomena observed in the physical environment. The subject builds on the competencies introduced at Junior Secondary School level under the learning area of Integrated Science. In Kenya Vision 2030 and sessional papers No. 1 of 2005 and No. 1 of 2019, the importance of science, technology and innovation has been emphasised for human capital development through education and training. Physics is an integral part of the STEM subjects in the Basic Education Curriculum Framework. It employs a scientific methodology of study with an emphasis on experimental approach to investigation, enhancing understanding of fundamental scientific concepts and principles. This leads to the acquisition of knowledge, skills, attitudes and values through precise and accurate scientific processes. Physics provides the learner with opportunities to develop competencies by empowering them to be creative and innovative, leading to independent approaches in problem solving and management of their environment. It also prepares learners for further training and the world of work by providing careers in STEM-related pathways. The content will be anchored on Kolb Theory of Experiential Learning and Constructivism Theory for teaching and learning.

GENERAL LEARNING OUTCOMES

By the end of Senior School the learner should be able to:

1. Relate Physics to technology and society to enhance the learner's appreciation of the physical environment. 2. Develop science process skills among learners through the use of appropriate instruments as they discover and explain the order of the physical environment.
3. Apply basic research and scientific skills to manipulate the environment and solve human problems. 4. Develop capacity for critical thinking through basic scientific skills and research in addressing pertinent and contemporary issues affecting the society.
5. Apply the principles of Physics for enhancement of innovations and entrepreneurial skills for development. 6. Use

relevant skills and values to promote local and global citizenship for harmonious coexistence and appreciation of diversity in people.

7. Acquire adequate knowledge, skills, values and attitudes to enhance exploitation of individual talents for leisure, self fulfillment, career growth, and for further education and training.
8. Apply acquired knowledge, skills, values, and attitudes for effective communication and utilisation of information in technological advancement.

SUMMARY OF STRANDS AND SUB STRANDS

Strand	Sub Strand	Suggested Number of Lessons

1. Mechanics and Thermal Physics	1.1 Introduction to Physics	7
	1.2 Pressure	19
	1.3 Mechanical Properties of Materials	12
	1.4 Temperature and Thermal Expansion	12
	1.5 Moments and Equilibrium	18
	1.6 Energy, Work, Power and Machines	20
2.0 Waves and Optics	2.1 Properties of Waves	19
	2.2 Radioactivity and Stability of Isotopes	19
3.0 Electricity and Magnetism	3.1 Electrostatics	12
	3.2 Current Electricity	22
	3.3 Introduction to Electronics	7
4.0 Environmental and Space Physics	4.1 Greenhouse Effect and Climate Change	6
	4.2 Introduction to Space Physics	7
Total Number of Lessons		**180**

Note: The suggested number of lessons per Sub Strand may be less or more depending on the context.

STRAND 1.0 MECHANICS AND THERMAL PHYSICS

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

1.0 Mechanics and Thermal Physics	**1.1 Introduction to Physics** **(7 lessons)** • Meaning of Physics as a body of knowledge in science. • Branches of Physics as a field of study (*mechanics, electricity & magnetism, thermodynamics, geometrical optics, waves, electronics, modern physics, astronomy*). • Importance of Physics in day-to-day life. • Relationship of Physics	By the end of the sub strand, the learner should be able to: a) explain Physics as a body of knowledge in science, b) describe the branches of Physics as a field of study, c) outline the importance of physics in day-to-day life, d) relate Physics to other fields of study, e) identify possible career opportunities in the field of Physics, f) appreciate the importance of Physics in day-to-day life.	The learner is guided to: ● work with others to search for the meaning of Physics as a branch of science, ● discuss with peers the main branches of Physics, ● discuss with peers the importance of Physics in day-to-day life and share the findings with the class, ● discuss with peers the relationship of Physics with other fields of study, ● engage resource person(s) or use print or non-print media to search for information on career opportunities in the field of Physics, ● design, produce and present	How is Physics relevant in day to-day life?

	with other fields of study. • Possible career opportunities beyond Senior School.		career charts highlighting areas related to Physics.	

Core competencies to be developed • Communication and collaboration: teamwork is embraced as the learner participates and contributes to the group decisions while discussing the main branches of Physics and its importance in day-to-day life. • Learning to learn: the learner practices the sharing of learnt knowledge while engaging the resource person(s) in searching for information on career opportunities in the field of Physics. • Digital literacy: the learner interacts with digital technology by use of digital devices to effectively accomplish the task of searching for information on career opportunities in the field of Physics.
Values: • Responsibility: the learner plays different roles while designing, producing and presenting career charts and highlighting areas related to Physics. • Respect: the learner demonstrates open-mindedness and appreciates diverse opinions while presenting their discussion findings on the importance of Physics in day-to-day life and related career opportunities.
Pertinent and Contemporary Issues (PCIs): Gender Disparity: the learner deals with myths and misconceptions about Physics as they engage resource persons or use print or non-print media to search for information on career opportunities in the field of Physics.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

1.0 Mechanics and Thermal Physics	**1.2 Pressure** **(19 lessons)** • Atmospheric pressure and its effects. • Factors affecting pressure in liquids. • Application of equation $P = \rho gh$ to determine pressure in liquid. • Transmission of pressure in fluids. • Applications of pressure in fluids: (*drinking*	By the end of the sub strand, the learner should be able to: a) describe atmospheric pressure as used in Physics, b) demonstrate the existence of atmospheric pressure in nature, c) investigate factors affecting pressure in fluids, d) apply the equation $P = \rho gh$ to determine pressure in fluids, e) demonstrate transmission of pressure in fluids, f) appreciate the applications of atmospheric pressure	The learner is guided to: ● discuss with peers the meaning of atmospheric pressure. ● carry out activities to demonstrate the existence of atmospheric pressure in nature, ● carry out activities to investigate and demonstrate factors affecting pressure in fluids, ● carry out experiments to derive and use the equation $P = \rho gh$ to determine pressure in liquid, ● discuss with peers the transmission of pressure in fluids, ● carry out activities to demonstrate the principle of transmission of pressure in	How do density of fluid, acceleration due to gravity and depth below the free surface affect pressure in fluid?

	straw, syringe, syphon, hydraulic machines and bicycle pump). • Mechanisms of water pumping.	and transmission of pressure in fluids in day-to-day life.	fluids and relate with the fluid pressure formula, ● visit the nearby garage, observe working of braking system (functioning of parts; braking) and the related concept to applications for safety, ● discuss with peers the relationship between air pressure of car tyres and the loading capacity of the vehicle for safety of the road users, ● discuss with peers the applications of atmospheric pressure and transmission of pressure in fluids in day-to day life, ● use print or non-print media to search for more information on the applications of atmospheric pressure and	

			transmission of pressure in fluids in day-to day life.	

Core competencies to be developed ● Communication and collaboration: the learner develops speaking and listening skill as they discuss the applications of atmospheric pressure and transmission of pressure in fluids. ● Digital Literacy: the learner acquires manipulation skills as they use digital media to search for more information on the applications of atmospheric pressure and transmission of pressure in fluids.
Values ● Love: the learner practices the virtue of sharing as they use available resource and avoid inflicting pain on others while carrying out activities to demonstrate the existence of atmospheric pressure in nature. ● Respect: the learner shows respect for diverse opinions as they discuss the transmission of pressure in fluid
Pertinent and Contemporary Issues (PCIs): Environmental Issues: • Climate change: the learner relates the effect of atmospheric pressure to climate change as they discuss the applications of atmospheric pressure and transmission of pressure in fluids. Social-Economic Issues • General History of Africa (GHA): as the learner examines the applications of pressure in fluids in water transport due to change in wind patterns, which facilitated Indian Ocean and Trans-Atlantic Trade.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.3 Mechanical Properties of Materials** **(12 lessons)** • Properties of materials (*ductility, malleability, elasticity, brittleness, strength, hardness, stiffness*). • Law of elasticity (Hooke's Law). • Stress, strain and modulus of elasticity. • Industrial	By the end of the sub strand, the learner should be able to: a) explain the mechanical properties of materials, b) demonstrate the mechanical properties of materials, c) determine the tensile stress and strain using mathematical formulae, d) describe applications of mechanical properties of materials, e) appreciate the	The learner is guided to: ● discuss with peers the mechanical properties of locally available materials, ● carry out activities to demonstrate the mechanical properties (*ductility, malleability, elasticity, brittleness, strength, hardness, stiffness and any other relevant and appropriate property*) of locally available materials, ● carry out activities to determine the relationship between tensile force and extension to illustrate mechanical properties of	1. Why does a string snap easily as compared to a spring? 2. Why is it important to study the mechanical properties of materials?

	applications of the properties of materials.	importance of knowledge on mechanical properties of materials in day today life.	materials: *constant of elasticity, tensile stress, breaking stress, tensile strain and modulus of elasticity as used in different materials,* ● discuss with peers the applications of Hook’s Law in car shock absorbers springs, ● use print and non-print media to search for applications of various mechanical properties of materials, ● use mathematical relationships to determine tensile stress, tensile strain and modulus of elasticity of materials. *Stress=F/A* *Strain* =Δ��/���� �� =	

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Core competencies to be developed:

● Communication and collaboration: the learner develops speaking skills as they discuss the mechanical properties of locally

available materials.

● Critical thinking and problem solving: the learner develops the skill of interpretation and inference as they explore problems by creating different possible solutions and their pros and cons while using mathematical relationships to determine tensile stress, tensile strain and modulus of elasticity of materials.

Values

● Responsibility: the learner shows accountability as they take care of apparatus used to perform experiments while determining the constant of elasticity, tensile stress, tensile strain and modulus of elasticity of material. ● Respect: the learner shows etiquette and humility during group discussions on mechanical properties of materials and their relevance in day-to-day life.

● Integrity: the learner shows virtue of honesty as they present genuine results that reflect the actual findings and observations of group discussions and practical activities on mechanical properties of materials.

Pertinent and Contemporary Issues (PCIs):

Socio-economic and environmental issues: the learner acquires safety and security skills as they appreciate the importance and application of mechanical properties of materials in making informed choices of materials for specific purposes.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.4 Temperature and Thermal Expansion** **(12 lessons)** • Temperature and its units. • Thermal expansion (*Linear expansivity, unusual expansion of water*). • Applications of thermal expansion. • Measurement of temperature (*liquid expansion devices, bimetallic devices,*	By the end of the sub strand, the learner should be able to: a) explain the meaning of temperature as used in thermal physics, b) measure temperature using different technologies, c) investigate thermal expansion and contraction in solids and fluids, d) describe applications of thermal expansion in solids and	The learner is guided to: ● discuss with peers the meaning of temperature, ● carry out activities to measure temperature using different temperature measurement technologies (*liquid expansion devices, bimetallic devices, thermocouples, resistive temperature devices (RTDs, thermistors), infrared radiators,*	1. Why is the lid of a sufuria made wider? 2. Why does a glass bottle break when the water in it freezes?

	thermocouples,	fluids, e) appreciate the applications of	*molecular change-of-state and silicon diodes, motion sensors*), ● use digital media to	

	resistive temperature devices (RTDs, thermistors), infrared radiators, molecular). • Change-of-state and silicon diodes.	thermal expansion in day-to-day life.	search for more information on measuring temperature using different temperature measurement technologies (*liquid expansion devices, bimetallic devices, thermocouples, resistive temperature devices (RTDs, thermistors), infrared radiators, molecular change-of state and silicon diodes, motion sensors*), ● carry out activities to demonstrate thermal expansion and contraction in solids	

			and determine linear expansivity of metals (*iron, steel, copper and*	

			aluminium wire), ● perform experiments to demonstrate thermal expansion and contraction in fluids (*include unusual expansion of water*), ● discuss the applications of thermal expansion in day-to-day life (*thermostat used in electrical devices, flash light/indicator system, construction industries, power lines, bridges, metal work and any other*), ● search for more information from the print or non-print media on the applications of	

			thermal expansion	

			(*thermostat used in electrical devices, flash light/indicator system, construction industries, power lines, bridges, metal work and any other*).	

Core competencies to be developed:

• Citizenship: the learner develops ethical digital citizenship skills as they search, access and use internet-based services when searching for more information from the print or non-print media on the applications of thermal expansion. • Digital literacy: the learner develops manipulative skill of interacting with digital technology as they use digital technology to effectively accomplish own tasks while searching for information from the print or digital media on the applications of thermal expansion and measurement of temperature.

Values:

• Peace: the learner shows love by respecting others and their opinions while discussing applications of thermal expansion in day-to-day life.

• Patriotism: the learner shows dedication by being conscious of their social and moral duties as they carry out activities to demonstrate thermal expansion and contraction in solids and determine linear expansivity of metals.

Pertinent and Contemporary Issues (PCIs) Socio-economic and environmental issues: the learner acquires safety and security skills by being keen on fire safety while performing experiments to demonstrate thermal expansion and contraction in fluids.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

1.0 Mechanics and Thermal Physics	**1.5. Moments and equilibrium** **(18 Lessons)** • C.O.G and Stability. • Principle of moments. • Moment of a force. • Principle of moment. • Moment about one point of support. • Torque and Couple. • Moment about two points of support. • Resolution of forces.	By the end of the sub strand the learner should be able to: a) determine the Centre of gravity of regularly and irregularly shaped objects, b) identify the states of equilibrium in bodies, c) determine moment of force about a point, d) verify the principle of moments in turning of objects, e) describe torque and couple in turning objects, f) appreciate the applications of moments and stability in day-to-day life.	The learner is guided to: ● design and carry out activities to determine the position of centre of gravity of regularly and irregularly shaped objects, ● carry out activities to demonstrate the stability, instability and neutral states of equilibrium of objects, ● discuss the meaning of moments of a force, ● carry out activities to demonstrate the turning effect of forces about a point, ● carry out activities to demonstrate moment about two points of support. ● carry out activities to	How does the stability of bodies affect the designs of their structures?

	• Applications of moments.		demonstrate and determine resolution of forces, ● carry out activities to demonstrate torque and couple, ● use mathematical relationships to determine centre of gravity and moments, ● carry out activities to investigate the factors that affect stability of objects, ● use print or non-print media to search for the applications of torque, couples and stability of bodies, ● take a walk to a nearby shopping centre, identify a	

			vehicle, discuss with peers the relationship between stability of the vehicle and its load carrying capacity.	

Core competencies to be developed

- Creativity and Imaginations: The learner develops observation skills as they discover fresh ways of doing things while

carrying out activities to demonstrate torque and couple.
- Citizenship: the learner develops active community life skills as they show responsibility to the community while carrying out activities to investigate the factors that affect stability of objects.

Values

- Unity: as the learner shows fairness by displaying team spirit while designing and carrying out activities to determine the position of centre of gravity of regular and irregular objects.
- Integrity: the learner practises the virtue of honesty while presenting genuine results as they design and carry out activities to determine the position of centre of gravity of regular and irregular objects.

Pertinent and contemporary issues (PCIs)

Socio-economic and environmental issues: the learner develops safety and security skills as they become conscious of road safety while carrying out activities to investigate the factors that affect stability of objects.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Mechanics and Thermal Physics	**1.6 Energy, Work, Power and Machines** **(20 Lessons)** • Kinetic energy, potential energy (*elastic and gravitational potential energy*) and law of conservation of energy. • Work and power. • Machines (*levers, inclined plane, pulleys, wheel and axle, gears,*	By the end of the sub strand the learner should be able to: a) explain the meaning of energy, work and power in relation to machines, b) demonstrate the transformation of mechanical energy using simple apparatus, c) describe applications of simple machines in making work easier, d) appreciate the applications of machines in day-to-day life.	The learner is guided to: • discuss with peers and explain the meaning of the terms (*energy, work, power and machines*) as used in physics, • carry out activities to demonstrate the concepts of energy, work, power and machines, • perform experiments to demonstrate mechanical energy transformations and forms of potential energy (potential energy to kinetic energy and back, *gravitational potential energy and elastic potential energy*), • discuss with peers, mechanical energy transformations and	How do machines make work easier?

	hydraulic lift, pulley belt, the		forms of potential energy (*potential energy to kinetic*	

	screw). • Applications of Machines; also include treadmill, elevators, escalators, excavator.		*energy and back, gravitational potential energy and elastic potential energy*), • visit a nearby garage, observe and identify functioning components of a motor vehicle, related energy transformations and necessary safety required when using the vehicle(*loud sound, bright light, safety belts*), • discuss with peers, energy transformations as a result moving parts of a motor vehicle's parts (*mechanical energy, chemical energy, electrical energy, kinetic energy, potential energy, heat energy and sound energy*), • apply mathematical relationships to deduce numerical tasks involving	

			kinetic energy, potential energy, work, power, mechanical advantage, velocity ratio and efficiency of simple machine, • demonstrate the law of	

			conservation of mechanical energy (*using a swinging pendulum, ball thrown upwards, a catapult, bow & arrow*), • use print or non-print media to search for information on the transformations of mechanical energy and applications of various simple machines in day to-day life (*levers, inclined plane, pulleys, wheel and axle, gears, hydraulic lift, pulley belt, screw),* • use print or non-print media to search for information on the use of simple machines in the construction of treadmills, elevators, escalators, among others. • use locally available materials to construct various simple machines.	

Core competencies to be developed • Communication and collaboration: the learner develops team spirit and speaking skills while discussing with peers in explaining the meaning of the terms used in Physics. • Critical thinking and problem solving: the learner develops the evaluation and decision making skill while applying mathematical relationships to deduce numerical tasks involving energy, work, power, and machines in day-to-day life (*velocity ratio, mechanical advantage & efficiency*).
Values • Love: the learner portrays a caring attitude while working with others to construct simple machines. • Responsibility: as the learner shows accountability by caring for own property and those of others while performing experiments to demonstrate mechanical energy transformations and forms of potential energy.
Pertinent and Contemporary Issues (PCIs): Life skills and values education • Socio-Economic and Environmental Education: the learner appreciates environment and technology while performing experiments to demonstrate mechanical energy transformations and forms of potential energy. • General history of Africa: as the learner appreciates the development of machines from simple to complex machines / evolution of machines from archaic to modern (e.g. steam engine to electric engines).

Suggested Assessment Rubric

Level Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations

Ability to explain the importance of Physics and possible career opportunities related to it.	Correctly and comprehensively explains the importance of Physics and possible career opportunities related to it.	Correctly explains the importance of Physics and possible career opportunities related to it.	Partially explains the importance of Physics and possible career opportunities related to it.	Explains the importance of Physics and some career opportunities related to it with hints.
Ability to determine pressure in solids and fluids.	Correctly and consistently determines pressure in solids and fluids.	Correctly determines pressure in solids and fluids.	Partially determines pressure in solids and fluids.	Determines pressure in solids and fluids with prompts.
Ability to explain the applications of pressure in solids and fluids.	Correctly and comprehensively explains the applications of pressure in solids and fluids.	Correctly explains the applications of pressure in solids and fluids.	Partially explains the applications of pressure.	Explains the applications of pressure in solids and fluids with cues.
Ability to determine tensile stress, strain and Young's modulus in materials.	Correctly and systematically determines tensile stress, strain and Young's modulus in materials.	Correctly determines tensile stress, strain and Young's modulus in materials.	Determines tensile stress, strain and Young's modulus in materials omitting some	Determines tensile stress, strain and Young's modulus in materials with cues.

			steps.	
Ability to describe the	Correctly and	Correctly describes the	Partially describes	Describes the

Level Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
applications of mechanical properties of materials.	comprehensively describes the applications of mechanical properties of materials.	applications of mechanical properties of materials.	the applications of mechanical properties of materials.	applications of mechanical properties of materials with prompt.
Ability to investigate thermal expansion and contraction in solids and fluids.	Correctly and systematically investigates thermal expansion and contraction in solids and fluids.	Correctly investigates thermal expansion and contraction in solids and fluids.	Investigates thermal expansion and contraction in solids and fluids, leaving out some steps.	Investigates thermal expansion and contraction in solids and fluids with help.

Ability to describe applications of thermal expansion and contraction in solids and fluids.	Correctly and comprehensively describes applications of thermal expansion and contraction in solids and fluids.	Correctly describes applications of thermal expansion and contraction in solids and fluids.	Partially describes applications of thermal expansion and contraction in solids and fluids.	Describes applications of thermal expansion and contraction in solids and fluids with prompt.
Ability to verify the principle of moments in turning of objects.	Correctly and systematically verifies the principle of moments in turning of objects.	Correctly verifies the principle of moments in turning of objects.	Verifies the principle of moments in turning of objects omitting some steps.	Verifies the principle of moments in turning of objects with prompt.
Ability to describe the applications of	Correctly and comprehensively	Correctly describes the applications of	Partially describes the applications of	Describes the applications of

Level Indicator	**Exceeds Expectations**	**Meets Expectations**	**Approaches Expectations**	**Below Expectations**
moments and stability in day-to-day life.	describes the applications of moments and	moments and stability in day to-day life.	moments and stability in day-to day life.	moments and stability in day-to-day life with

	stability in day-to-day life.			prompts.
Ability to demonstrate the transformation of mechanical energy using simple apparatus.	Correctly and systematically demonstrates the transformation of mechanical energy using simple apparatus.	Correctly demonstrates the transformation of mechanical energy using simple apparatus.	Demonstrates the transformation of mechanical energy using simple apparatus omitting some steps.	Demonstrates the transformation of mechanical energy using simple apparatus, with prompt.
Ability to describe applications of simple machines in making work easier.	Correctly and comprehensively describes applications of simple machines in making work easier.	Correctly describes applications of simple machines in making work easier.	Partially describes applications of simple machines in making work easier.	Describes applications of simple machines in making work easier with cues.

STRAND 2.0 WAVES AND OPTICS

Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Suggested Key Inquiry Question(s)**

2.0 Waves and Optics	**2.1 Properties of Waves** **(19 Lessons)** • Rectilinear propagation, reflection, refraction, diffraction, interference. (*Qualitative treatment only*). • Applications of properties of waves (*Need for modulation. Production and detection of frequency modulated wave*).	By the end of the sub strand, the learner should be able to: a) explain the wave properties in real-life situations, b) demonstrate the properties of waves in nature, c) demonstrate the formation and properties of stationary waves in nature, d) describe applications of stationary waves in day to-day life, e) describe Doppler effect and its applications in day-to-day life, f) appreciate formation	The learner is guided to: ● discuss with peers the meaning of wave properties and their applications in day-to-day life, ● perform experiments with peers to demonstrate wave properties, sketch wave patterns and make presentation (*interference of waves*) Note: ***Qualitative treatment only,*** ● perform experiments to demonstrate formation and properties of stationary waves, citing their respective applications and associated numerical relationships in modes of vibration	1. How do you relate waves to the basic properties of light? 2. Where is Doppler Effect applied in real life situations?

	• Stationary waves and its applications (*Resonance*). • Doppler Effect and applications.	and application of waves in real-life situations.	(*vibrating strings, vibrating air columns, resonance, detection of frequency modulated waves*), ● use print or non-print media to search for information on the behaviour of sound waves from an approaching and receding ambulance or music from moving vehicles and discuss the variations in the frequencies of sound heard to describe Doppler effect (Qualitative treatment only).	

Core competencies to be developed:

- Learning to learn: the learner practises the act of exchanging information while performing experiments with peers to demonstrate wave properties, sketch wave patterns and make presentation.
- Self-efficacy: the learner develops leadership skills while ordering and prioritising tasks as they perform experiments to demonstrate formation and properties of stationary waves, citing their respective applications and associated numerical relationships in modes of vibration.

Values • Social Justice: The learner shows cooperation while according each other equal opportunities in sharing responsibilities while performing experiments with peers to demonstrate wave properties, sketch wave patterns and make presentation

• Patriotism: The learner shows dedication by being conscious of their social and moral duties while using print or non-print media to search for information on the behaviour of sound waves from an approaching and receding ambulance or music from moving vehicles and discuss the variations in the frequencies of sound heard to describe Doppler Effect.
Pertinent and Contemporary Issues (PCIs): Socio-economic and environmental education: The learner acquires environmental conservation skills as they prevent pollution by noise while performing experiments to demonstrate formation and properties of stationary waves, citing their respective applications and associated numerical relationships in modes of vibration.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

2.0 Waves and Optics	**2.2 Radioactivity and Stability of Isotopes** **(19 lessons)** • Stability of isotopes. • Radioactive decay. • Types and properties of radiations. • Safety precautions • Detection of radiations. • Half-life and its significance. • Nuclear fission, nuclear fusion and significance. • Chain reaction and	By the end of the sub strand, the learner should be able to: a) explain the terminologies used in radioactivity, b) identify the types and properties of radioactive emissions in nature, c) illustrate how radionuclides attain their stability using nuclear equations, d) demonstrate the detection of radioactive emissions in nature, e) determine the half-life ($t_{1/2}$) of common radioactive elements, f) appreciate the	The learner is guided to: ● search and discuss with peers the meaning of terms used in radioactivity (*nuclear stability, radioactivity, half-life, nuclide, nucleotides, radioactive decay radioisotope, background radiation*), ● discuss with peers the types and properties of radiations (*nature, electrical charge, relative mass, absorption, velocity, ionising power, effect in electric and magnetic fields*), ● make charts that illustrate properties of radioactive radiations (*penetration power, ionising effect, behaviour in electric and magnetic fields*) ● discuss with peers and write nuclear equations to illustrate how	1. How is radioactivity important in day-to-day life? 2. What are the risks of exposure to radiation?

	nuclear equations. • Medical and industrial applications of radioactivity.	applications and safety precautions of radioactivity in real life.	radionuclides attain stability, ● use charts showing chain reactions to identify different radioactive emissions and write resultant nuclear equations (*a uranium nuclide natural decay series*) ● search, discuss in groups and present findings on safety precautions when handling and disposing of radioactive substances, ● work in groups to demonstrate the detection of radioactive emissions by photographic emulsion/plate, cloud chamber, leaf electroscope, *Geiger muller* using print and non print media, ● carry out activities with peers to demonstrate half-life ($t_{\frac{1}{2}}$) (*timing running water from a burette*),	

			● use data sourced from print or non print media, to determine the half life of common radioactive elements using formula:	

			$N = No\left(\frac{1}{2}\right)^{\frac{T}{t_{\frac{1}{2}}}}$ and graphical method (*decay curve*), ● discuss with peers qualitative treatment of nuclear fission and fusion (mention nuclear reactions as a source of energy **NB:** Nuclear reactions are different *from chemical reactions*), ● search from print and non-print media, applications of radioactivity in day-to-day life (*chemistry, medicine, industry, carbon dating and agriculture*) and	

			dangers (*safety in hospitals*, *pollution e.g. the Chernobyl disaster)*	

Core competencies

- Imagination and Creativity: the learner develops the skill of making connections by finding hidden patterns between different ideas while making charts that illustrate properties of radioactive radiations.
- Self-efficacy: the learner develops self-awareness and acquires planning skill while searching and discussing in groups safety precautions when handling radioactive substances.

Values

- Respect: the learner shows acceptance as they understand and appreciates others when searching, discussing in groups and presenting findings on safety precautions when handling radioactive substances.
- Peace: the learner shows care by avoiding to hurt others while working with peers to demonstrate the detection of radioactive emissions.

Pertinent and Contemporary Issues (PCIs):

- Citizenship education: the learner practises peace and conflict resolution mechanisms as ways of promoting peace while searching from print and non-print media, on applications of radioactivity.
- Socio-Economic and Environmental issues: the learner demonstrates environmental conservation awareness as they search, discuss in groups and present findings on safety precautions when handling and disposing of radioactive substances.

Suggested Assessment Rubric

Level Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to describe the formation, properties and effects of waves.	Correctly and comprehensively describes the formation, properties and effects of waves.	Correctly describes the formation, properties and effects of waves.	Partially describes the formation, properties and effects of waves.	Describes the formation, properties and effects of waves with prompt.
Ability to illustrate how radionuclides attain their stability using nuclear equations.	Correctly illustrates how radionuclides attain their stability using nuclear equations.	Correctly illustrates how radionuclides attain their stability using nuclear equations.	Partially illustrates how radionuclides attain their stability using nuclear equations.	Illustrates how radionuclides attain their stability using nuclear equations with hints.
Ability to describe the industrial applications and safety precautions of radioactivity in real life.	Correctly and comprehensively describes the industrial applications and safety precautions of radioactivity in real life.	Correctly describes the industrial applications and safety precautions of radioactivity in real life.	Partially describes the industrial applications and safety precautions of radioactivity in real life.	Describes the industrial applications and safety precautions of radioactivity in real life with prompt.

STRAND 3.0 ELECTRICITY AND MAGNETISM

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions

3.0 Electricity and Magnetism	**3. 1 Electrostatics** **(12 Lessons)** • Electric field patterns. • Force between charges. • Charge distribution on conductors. • Applications of electrostatics.	By the end of the sub strand the learner should be able to; a) explain the origin of charges in a material, b) describe methods of charging a conductor, c) illustrate charge distribution on conductors of various shapes, d) explain functions of the various parts of an electroscope, e) illustrate charging of an electroscope, f) appreciate uses of an electroscope and applications of static charges in day-to-day life.	The learner is guided to: ● discuss with peers the origin of charges on materials, (*atom, nucleus, neutrons, protons and electrons), SI unit of charge and the law of electrostatics*), ● perform experiments to demonstrate the generation of static charges on bodies (*through rubbing*), ● discuss with peers on various ways of charging a conductor *(contact, induction and separation)* while sketching the distribution of charges, ● describe the features and functions, charging and discharging/earthing of a leaf electroscope (*contact and induction*), ● construct a simple leaf	1. How do lightning arrestors work? 2. How do materials get charged?

			electroscope using locally available materials, ● perform experiments to verify the uses of an electroscope in electrostatics (*testing for presence, type, and quantity of charge as well as conduction and insulation properties of materials*), ● use print and non-print media to investigate the distribution of charges on metallic conductors (*spherical, wedge shaped, pear-shaped and sharp conductor among others*), ● engage resource persons or use print and non-print media to explain the applications of electrostatics and real-life effects of electrostatics (*spray gun, photocopiers, finger printing, electrostatic precipitators, lightning*	

			and associated safety measures and many more).	

Core competencies to be developed: • Learning to learn: the learner practises how to organise and learn independently while constructing a simple leaf electroscope using locally available materials. • Self-efficacy: the learner develops the skill of effective communication by executing tasks while performing experiments to verify the uses of an electroscope in electrostatics.
Values: • Responsibility: the learner shows self-drive as they engage in assigned roles and duties while constructing a simple leaf electroscope. • Love: the learner shows care by respecting others during group discussions with peers on the origin of charges on materials.
Pertinent and Contemporary Issues (PCIs): Safety and Security: the learner shows awareness of safety in the class and school environment as they engage resource persons or use print and non-print media to find out the applications of electrostatics and real-life effects of electrostatics.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Question(S)
3.0 Electricity and Magnetism	**3.2 Current Electricity** **(22 lessons)** Measurement of current, a.m., potential difference and resistance. • Investigate the relationship between current, voltage and resistance for various materials. • Determine resistance by resistor colour codes. • Laws of current and voltage in a	By the end of the sub strand, the learner should be able to: a) explain the terminologies used in current electricity, b) verify the relationships: *$V=IR$, $E=I(R+r)$* as used in current electricity, determine the resistance and resistivity of various conductors using different methods, c) determine effective resistance of various resistor networks, d) determine the relationship of potential difference and current	The learner is guided to: • discuss the meaning of current, potential difference, electromotive force and internal resistance as used in current electricity, • perform an experiment to investigate the relationship between potential difference and current through a conductor and draw the requisite graphs to verify Ohm's law, • perform an experiment to investigate the relationship between e.m.f., potential difference, current, resistance through a conductor and internal resistance of a cell and draw the requisite graphs, • derive the mathematical	1. How is current electricity applicable in our day-to day life? 2. Why are resistors used in electrical circuits?

	circuit.	to power as used in		

	• Resistors and Resistor networks. • Applications of resistors. • Electrical heating and applications.	heating effect of an electric current, e) appreciate the applications of current electricity in day-to-day life.	relationship between emf, potential difference, electric current, resistance, internal resistance and use it to solve numerical tasks, • carry out an experiment to classify the types of resistors as either ohmic or non-ohmic, • perform experiments with peers to investigate the factors affecting resistance of ohmic resistors (*temperature, length, cross section area and type),* • perform experiments to determine resistance using various methods (*resistor colour codes, ammeter voltmeter method, wheatstone bridge and the metre bridge, resistor*	

			networks), • carry out experiments to determine the relationship of potential difference and	

			current to power (P=VI), • discuss in groups the applications of heating effect of electric current *(potential difference, current, resistance, time)* and classify them as either ohmic or non-ohmic, • search for information pertaining to the importance of resistors in day-to-day life from available sources and resources.	

Core competencies to be developed • Critical Thinking and Problem Solving: the learner develops the evaluation and decision-making skill while deriving the mathematical relationship between emf, potential difference, electric current, resistance and use it to solve related problems. • Learning to learn: the learner develops the skill of carrying out research while performing an experiment to investigate the relationship between potential difference and current through a conductor and draw the requisite graphs.
Values • Responsibility: as the learner takes accountability as they care for own property and those of peers while performing experiments with peers to investigate the factors affecting resistance of Ohmic resistors. • Respect: the learner shows communication etiquette and humility during discussions and practical sessions.

Pertinent and Contemporary Issues (PCIs): Socio-Economic and Environmental Issues (safety and security): the learner appreciates safety in the laboratory and the applications of electricity as they perform experiments to determine resistance using various methods.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

3.0 Electricity and Magnetism	**3.3 Introduction to Electronics** (7 lessons) • Energy band theory. • Insulators. • Conductors. • Semiconductors. • Intrinsic and extrinsic semiconductors. • Superconductors.	By the end of the sub strand, the learner should be able to: a) explain the meaning of insulator, conductor, semiconductor and superconductor, b) distinguish between insulators, conductors, semiconductors and superconductors, c) investigate the electrical behaviour of conductors, semiconductors and insulators with varying temperatures, d) explain intrinsic and extrinsic semiconductors as used in Physics, e) explain the formation of n-type and p-type	The learner is guided to: • discuss with peers the meaning of insulator, conductor, semiconductor, and superconductors, • perform an experiment to investigate the electrical behaviour of conductors, semiconductors and insulators with varying temperatures, • use diagrams to distinguish between conductors, semiconductors and insulators using energy band theory, • discuss the meaning of intrinsic and extrinsic semiconductors, • use print and non-print devices to search for	1. How does temperature affect the resistance of conductors and semiconductors? 2. What is the significance of semiconductors in day-to-day life?

		semiconductors from intrinsic semiconductors, f) describe the applications of conductors, semiconductors, insulators and superconductors in day-to-day life, g) appreciate the applications of conductors, semiconductors, insulators and superconductors in day-to-day life.	information on the formation of p-type and n-type semiconductors, • discuss with peers the applications of conductors, semiconductors, insulators and superconductors in day to-day life, • discuss with peers the applications of conductors and insulators in a car wiring system, • search for information on applications of conductors, semiconductors, insulators and superconductors from relevant sources and	

			resources.	

Core competencies to be developed:

- Critical thinking and problem solving: the learners demonstrate ability to analyse complex problems and logical reasoning while performing an experiment to investigate the electrical behaviour of conductors, semiconductors and insulators with varying temperatures.
- Digital literacy: the learner interacts with digital technology as they dismantle and assemble parts of digital devices while performing an experiment to investigate the electrical behavior of conductors, semiconductors, and insulators with varying temperatures.

Values:

- Social Justice: the learner demonstrates equity by according equal opportunities to others when sharing responsibilities while searching for information on applications of conductors, semiconductors, insulators and superconductors from relevant sources and resources
- Unity: the learner ensures inclusion by collaborating with others while working in a group to perform an experiment to investigate the electrical behaviour of conductors, semiconductors and insulators.

Pertinent and Contemporary Issues (PCIs)

Citizenship education: the learner acquires good governance skills by applying integrity and principles of leadership as they work together to search for information on applications of conductors, semiconductors, insulators and superconductors from relevant sources and resources.

Suggested Assessment Rubric

Level Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to explain the charging and uses of an electroscope.	Correctly and comprehensively explains the charging and uses of an electroscope.	Correctly explains the charging and uses of an electroscope.	Partially explains the charging and uses of an electroscope.	Explains the charging and uses of an electroscope with hints.
Ability to describe applications of electrostatic charges.	Correctly and comprehensively describes applications of electrostatic charges.	Correctly describes applications of electrostatic charges.	Partially describes applications of electrostatic charges.	Describes applications of electrostatic charges with prompt.
Ability to determine resistance using different methods and resistivity of a conductor.	Correctly and consistently determines resistance using different methods and resistivity of a conductor.	Correctly determines resistance using different methods and resistivity of a conductor.	Partially determines resistance using different methods and resistivity of a conductor.	Determines resistance using different methods and resistivity of a conductor, with hints.

Ability to describe the applications of current electricity in day-to-day life.	Correctly and comprehensively describes the applications of current electricity in day-to-day life.	Correctly describes the applications of current electricity in day-to-day life.	Partially describes the applications of current electricity in day-to-day life.	Describes the applications of current electricity in day-to-day life with cues.

Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Key Inquiry Question(s)**

4.0 Environmental and Space Physics	**4.1 Greenhouse Effect and Climate Change (6 Lessons)** • Greenhouse effect. • Role of the ozone layer. • Global warming. • Climate change.	By the end of the sub strand, the learner should be able to: a) explain the greenhouse effect and climate change in the environment, b) outline the factors leading to greenhouse effect in the environment, c) explain the effect of ozone layer on climate change, d) describe the mitigating factors against climate change in the environment, e) appreciate the impact of climate change on the environment.	The learner is guided to: • discuss with peers the meaning of greenhouse effect and climate change and the terminologies associated (greenhouse gases, global warming, ozone layer), • discuss the factors leading to greenhouse effect (use examples such as; *emissions from vehicles temperatures within the greenhouse, cars packed with closed windows in the sun to describe the greenhouse effect*), • demonstrate the effects of climate change in the immediate environment (*level of water in the lakes, volume of rivers, vegetation change, change in weather patterns and changes in land use*),	1. How do human actions impact climate change? 2. How does ozone layer depletion threaten our environment?

			• discuss the role of human activities in escalating environmental degradation, • using reference materials, resource persons or digital materials, outline mitigating factors against climate change, • use print and non-print media to search for information that describes the greenhouse effect, global warming, ozone layer, and climate change.	

Core competencies to be developed:

- Communication and collaboration: the learner develops speaking skills while discussing with peers the meaning of greenhouse effect and climate change and related terminologies.
- Critical thinking and problem solving: the learner develops open-mindedness and creativity while exploring complex problems when researching and using reference materials, resource persons or digital materials, outline mitigating factors against climate change.
- Digital literacy: the learner interacts with digital technology while searching, and watching videos and animations to describe the greenhouse effect, global warming, ozone layer and climate change.

Values: • Peace: the learner shows respect for diverse opinions during discussions on greenhouse effect and climate change and related terminologies.

• Unity: the learner practises cooperation as they collaborate with others, while demonstrating the effects of climate change in the immediate environment.
Pertinent and Contemporary Issues (PCIs): Socio-economic and environmental education: The learner develops awareness on global warming as they use print and non print media to search for information that describes the greenhouse effect, global warming, ozone layer, and climate change.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Key Inquiry Question(s)

4.0 Environmental and Space Physics	**4.2 Introduction to Space Physics** **(7 lessons)** • Theory of the origin of the universe. • Evolution of astrophysics. • Classification of bodies in the universe. • Telescopy. • Planetary motion. • History of space exploration. • Careers in Space Flight.	By the end of the sub strand, the learner should be able to: a) describe the Big Bang Theory of the origin of the universe, b) classify celestial bodies in the universe, c) outline the evolution of astrophysics and space exploration, in space physics, d) explain the motion of planets around the sun, e) appreciate the careers in space exploration.	The learner is guided to: • discuss the Big Bang Theory on the origin of the universe, • use digital media to observe the celestial bodies, methods of space exploration and planetary motion, • model the planetary motion, • discuss with peers the methods of exploring the universe (*Telescopes*), • discuss with peers the careers available in astrophysics and space exploration.	1. How was the universe / earth formed? 2. How do we benefit from astrophysics?

Core competencies to be developed: • Digital literacy: the learner interacts and uses digital technology to accomplish own tasks to search for information and media on methods of space exploration and planetary motion.
Values to be developed • Integrity: the learner demonstrates fairness as they utilise the available resources prudently while modelling the planetary motion. • Love: the learner demonstrates self-sacrifice by putting the interest of others before own as they use digital media to observe the celestial bodies, methods of space exploration and planetary motion.
Pertinent and Contemporary Issues (PCIs) Self-management skills: the learner develops self-awareness as they discuss the Big Bang Theory on the origin of the universe (*Big bang theory*).

Suggested Assessment Rubric

Level Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectation	Below Expectation
Ability to describe the impact of climate change in the environment.	Correctly and comprehensively describes the impact of climate change in the environment.	Correctly describes the impact of climate change in the environment.	Partially describes the impact of climate change in the	Describes the impact of climate change in the environment with prompt.

			environment.	
Ability to explain the evolution of astrophysics and space exploration in space physics.	Correctly and comprehensively explains the evolution of astrophysics and space exploration in space physics.	Correctly explains the evolution of astrophysics and space exploration in space physics.	Partially explains the evolution of astrophysics and space exploration in space physics.	Explains the evolution of astrophysics and space exploration in space physics with hints.

APPENDIX: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Assessment Methods in Science	Learning Resources	Non-Formal Activities

● Reflections ● Game Playing ● Pre-Post Testing ● Model Making ● Explorations ● Experiments ● Investigations ● Conventions, Conferences and Debates ● Teacher Observations ● Project ● Journals ● Portfolio ● Oral or Aural Question(s)s ● Learner's Profile ● Written Tests ● Anecdotal Records	● Laboratory Apparatus and Equipment ● Textbooks ● Models ● Digital media (Radio and TV education programmes, Kenya Education Cloud and OERs) ● Print media (charts, pictures, journals, magazines) ● Digital Devices ● Software ● Recordings ● Resource persons	● Visiting the science historical sites. ● Using digital devices to conduct scientific research. ● Organising walks to have live learning experiences. ● Developing simple guidelines on how to identify and solve some community problems. ● Conducting science document analysis. ● Participating in talks by resource persons on science concepts. ● Participating in science clubs and societies. ● Attending and participating in science and engineering fairs. ● Organising and participating in exchange programmes. ● Making oral presentations and demonstrations on science issues.